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## CE 507: Advanced Concrete Science

**Course Code:** CE 507

**Course Name:** Advanced Concrete Science

L-T-P-C: 3-0-0-3

Prerequisite: CE 552 - Concrete Technology

Students intended for: UG/PG

Elective or Core: Elective

Approval: 9th Senate

### Course contents

- **Module I**

Mix design methods for ordinary and high performance concretes [6 Lectures]

- **Module II**

Performance in fresh and hardened states – Rheology, Stress-Strain characteristics, Creep and Shrinkage. [6 Lectures]

- **Pore structure of cement based materials**

Types and genesis of pores, Methods of measurement. [4 Lectures]

- **Transport processes in concrete**

Mechanisms of Water, Gas and Salt transport; Transport models; Tests for the assessment of transport characteristics. [6 Lectures]

- **Degradation mechanisms**

Carbonation, Sulphate attack, Freeze-Thaw, Alkali-Silica reaction. [6 Lectures]

- **Reinforcement corrosion**

Mechanisms of initiation and propagation, Electrochemical aspects, Monitoring, controlling and prevention, service-life prediction. [6 Lectures]

- **Design for Durability**

Service-life, Prescriptive, Performance and Model-based approaches. [6 Lectures]

### Text Books

1. Aligizaki, K. K., **Pore structure of cement-based materials: testing, interpretation and requirements**, Taylor & Francis, New York, 2005.
2. Richardson, M., **Fundamentals of durable reinforced concrete**, Spon Press, London and New York, 2003.
3. Hall, C. and Hoff, W.D., **Water transport in brick, stone and concrete**, CRC Press, Florida, 2011.

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**Reference Books:**

1. Sarja, A. and Vesikari, E., **Durability design of concrete structures-Report of RILEM technical committee 130-CSL**, E&FN Spon, London, 2004.
2. Cerny, R. and Rovnanikova, P., **Transport processes in concrete**, Spon Press, USA and Canada, 2002.
3. Claisse, P.A., **Transport properties of concrete: measurement and applications**, Woodhead Publishing, UK, 2014.
4. Bertolini, L., Elsener, B., Pedersen, P., Redaelli, E. and Polder, R.B., **Corrosion of steel in concrete: prevention, diagnosis, repair**, Wiley-VCH Verlag GmbH & Co. KGaA, Germany, 2013.
5. J. D. West and F. K. Levy, **A management guide to PERT/CPM with GERT/PDM/DCPM**, 1979.